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Palaeomagnetic Results from Palaeozoic Rocks of South-east Australia

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Summary

Palaeomagnetic results have been obtained from igneous rocks whose ages range from Middle Silurian (Wenlockian) to Lower Carboniferous (Viséan). A collection of 210 samples from four areas of south-east Australia was studied. The use of alternating field demagnetization procedures enabled the directions of presumed original magnetization to be isolated for the various formations. A conglomerate test provided additional stability evidence for the result from one formation. Calculated palaeomagnetic pole positions are as follows:

Isismurra Formation and Lower Kuttung Series (Viséan)—34°E, 73°N with $A_{95} = 21°$; Lochiel Formation (Upper Devonian)—140°E, 58°N with $A_{95} = 7°$; Ainslie Volcanics (Lower Devonian)—173°E, 71°N with $A_{95} = 10°$; Bowning Group (Gedinnian Stage, Lower Devonian)—225°E, 64°N with $A_{95} = 9°$; Laidlaw Formation, Douro Volcanics and Hawkins 'Series' (Wenlockian–Ludlovian, Middle–Upper Silurian)—91°E, 54°N with $A_{95} = 7°$.

1. Introduction

The palaeomagnetism of Palaeozoic rocks from Australia has been studied for a number of years. Irving & Green (1958) reported palaeomagnetic pole positions for the lower Palaeozoic following measurements of the natural remanent magnetization (NRM) of two formations from northern Australia. Green (1959, 1961) also studied the NRM of various Devonian and Silurian formations from south-east Australia. The introduction of methods for partial demagnetization of specimens enabled Briden to make an extensive investigation of many pre-Carboniferous formations from widespread areas of south-east Australia. He regarded most of the data as recording remagnetization during Late Carboniferous or Early Tertiary times (Briden 1965a, b). However, the Upper Silurian Mugga Mugga Porphyry appeared to have retained its original magnetization (Briden 1966). Robertson (1962) and Chamalaun (1968) reported preliminary results from Devonian formations. Detailed studies of the Upper Carboniferous (Irving 1966) and Permian (Irving & Parry 1963) produced much reliable information.

A review of the early studies of NRM indicated the existence of material warranting an examination using modern techniques. All previous data from the lower

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Palaeozoic were subsequently superseded by the results from a new collection of more than 300 samples of sediments and volcanics from northern Australia (Luck 1970; McElhinny & Luck 1970; Luck 1972). The early work had also suggested the presence of a stable primary magnetization in certain south-east Australian rocks. Of particular interest were the results recorded by Green (1961) for the Ainslie Volcanics and the Lochiel Formation (then termed the ' Yalwal Stage Basalts ') as well as for the Douro Volcanics (called the ' Duro Porphyry ' by Green 1959, quoted in Irving 1964). The success of demagnetization tests on a few specimens remaining from the original samples prompted new collections from the above three formations and other igneous rock units of south-east Australia.

Results from the four areas sampled (Fig. 1) will be described separately. The magnetizations of all specimens, cut as cylinders 2·8 cm × 2·8 cm, were measured using a spinner magnetometer (P.A.R. model SM 1). Generally, the NRMs of at least two specimens per sample were measured before applying alternating field (a.f.) and progressive thermal demagnetization techniques. The apparatus for magnetic

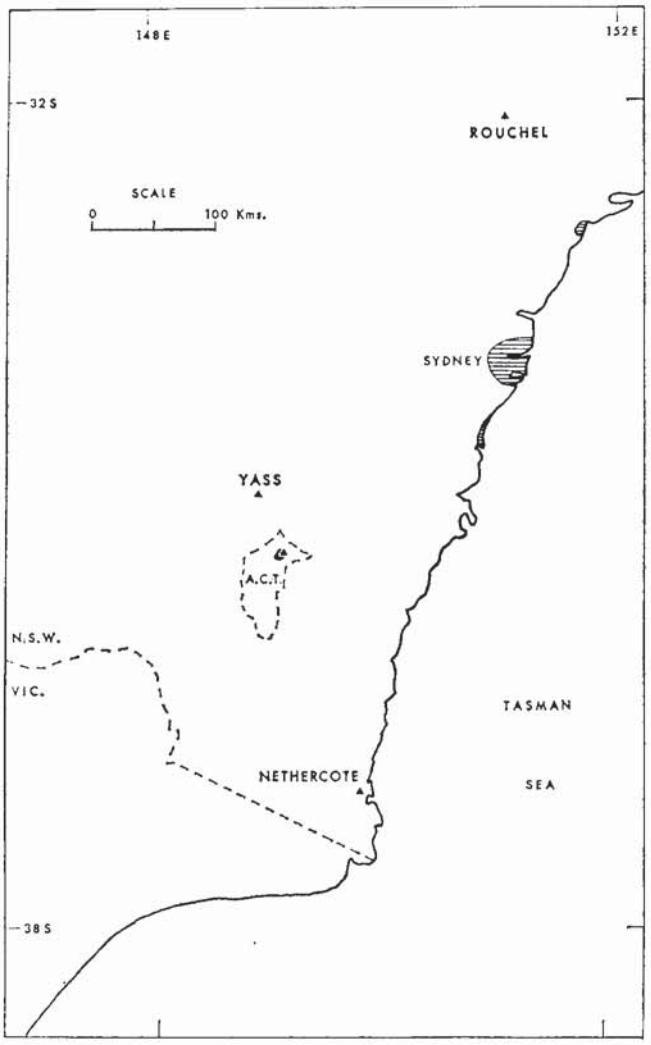


FIG. 1. Sketch map showing sampling localities in south-east Australia.

Table 1

Formations sampled in south-east Australia

Period	Unit	Sampling area	No. of samples	No. of specimens measured
Lower Carboniferous	Isismurra Formation	Rouchel	30	63
Upper Devonian	Lochiel Formation	Nethercote	48*	107
Lower Devonian	Ainslie Volcanics	A.C.T.	35	85
	Bowning Group	Yass	43	87
Upper Silurian	Laidlaw Formation		22	44
	Yass Group	Yass	10	51
	Douro Volcanics		18	38
Middle Silurian	Hawkins ' Series '		4	20

* Total includes nine samples collected by Green

cleaning was similar to the one described by McElhinny (1966), that for thermal cleaning being described by McElhinny, Luck & Edwards (1971).

The results have been analysed using the statistical method of Fisher (1953). The randomness of resultant directions has been tested at the 5 per cent probability level by comparing each vector resultant 'R' with the appropriate value tabulated by Watson (1956) or Vincenz & Bruckshaw (1960). All directions are referred to the palaeohorizontal by correcting for the structural attitude of the sampling sites.

Table 1 lists the locations, stratigraphic relationships, extent of sampling and number of specimens measured of the various rock units.

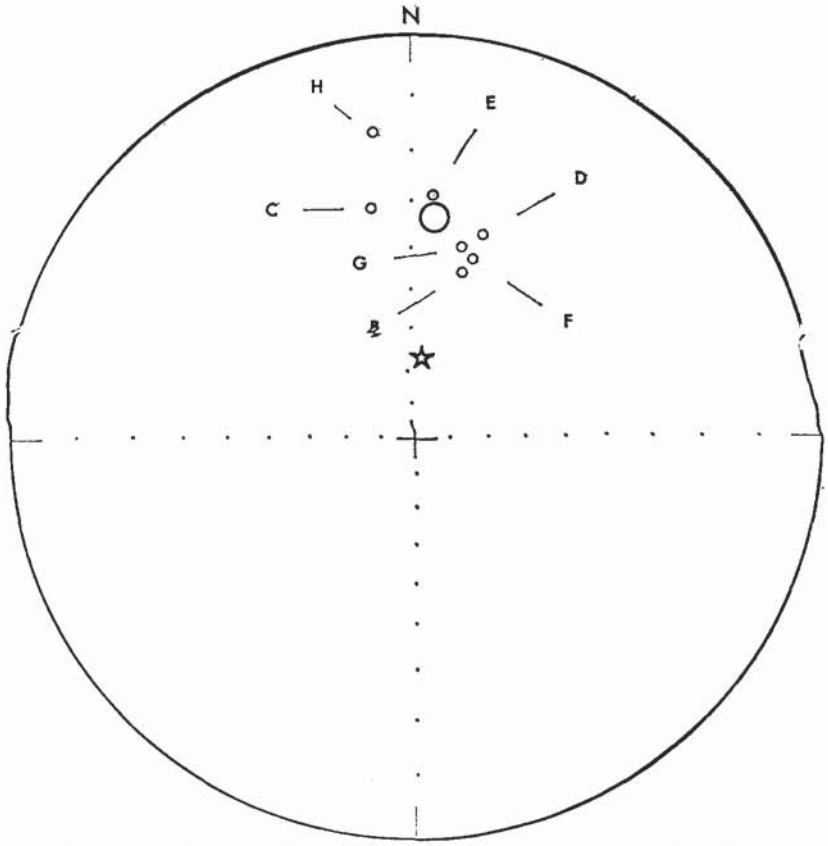


FIG. 2. Ainslie Volcanics. Mean cleaned directions for the seven sites with non-random magnetization after a.f. treatment. The large circle indicates the mean direction for the formation and the star represents the Earth's present field.

2. The Ainslie Volcanics

2.1 *Geology*

The Ainslie Volcanics consist of 210 m of interbedded acid to dacitic flows and pyroclastic rocks. They rest unconformably on the eroded surface of folded Silurian formations and appear to have been affected by the Tabberabberan Orogeny of the Middle Devonian (Öpik 1958). The Volcanics are therefore regarded as Early Devonian in age.

Mount Ainslie, the type locality, may be considered as the preserved core of ' the ruin of a volcano ' (Öpik 1958) and the bedded tuffs occurring on its slopes are mostly sub-horizontal.

2.2 *Sampling*

Thirty-two hand samples and three drilled cores were taken from two quarries and six freshly-exposed road cuttings on Mount Ainslie. The collection spanned a vertical distance of 170 m.

2.3 *Laboratory measurements and results*

The seven sites B–H produced samples with a very stable magnetization, the direction being practically unchanged by magnetic cleaning in peak alternating fields of 500 Oe. Representative specimens from these seven sites were heated to temperatures in the range 500–550°C with hardly any effect on their magnetization. Further heating rapidly reduced intensities and randomized directions of magnetization, thus suggesting the dominance of magnetite among the magnetic material present.

Table 2

Ainslie volcanics: site mean directions after cleaning

Site	$N(n)$	D	I	R	k	Pole (deg)	
A	3 (5)	**					
B	3 (6)	16	−42	2·99	406	204 E	72 N
C+	1 (4)	350	−27	(3·95)	(60)	122 E	68 N
D	7 (14)	19	−33	6·94	96	198 E	66 N
E	6 (12)	06	−26	5·96	130	164 E	68 N
F	3 (6)	19	−40	2·99	619	207 E	69 N
G	2 (4)	14	−37	1·99	3620	193 E	71 N
H	2 (4)	353	−15	1·99	520	133 E	62 N
Formation mean direction ($\alpha_{95} = 11°$)							
	7	07	−32	6·82	33·4		
Formation mean of site poles ($A_{95} = 10°$)							
				6·85	38·7	173°E	71°N

Intensities of NRM were in the range 0·2 to 130×10^{-6} emu cm^{-3}

Peak field normally used for cleaning was 500 Oe.

** = Direction random by criteria in text.

\+ = Specimens taken 8 cm apart along core.

N = Number of hand samples or cores.

n = Number of specimens measured.

D = Declination in degrees, east of north.

I = Inclination in degrees, positive downwards.

R = Resultant of N unit vectors.

k = Fisher's precision parameter

α_{95} A_{95} = Semi-angles of cones of 95 per cent confidence for mean direction and mean pole respectively.

Site A was situated at the summit of Mount Ainslie, and partial demagnetization of specimens from the two hand samples and one core did not reveal a stable direction of magnetization. Eight of the 14 samples collected from one of the quarries, site E, were discarded for the same reason.

The results after a.f. demagnetization are summarized in Fig. 2 and Table 2, which also show the formation mean direction of primary magnetization, $D = 07°E$, $I = -32°$. This is in quite good agreement with the direction of NRM obtained by Green ($D = 17°E$, $I = -30°$). The palaeomagnetic pole is situated at 173°E, 71°N with $A_{95} = 10°$.

3. The Lochiel Formation

3.1 *Geology*

The volcanics of the Lochiel Formation are chiefly basaltic and doleritic lava flows with minor rhyolites and felsites (Hall 1960). The flows, some showing advanced

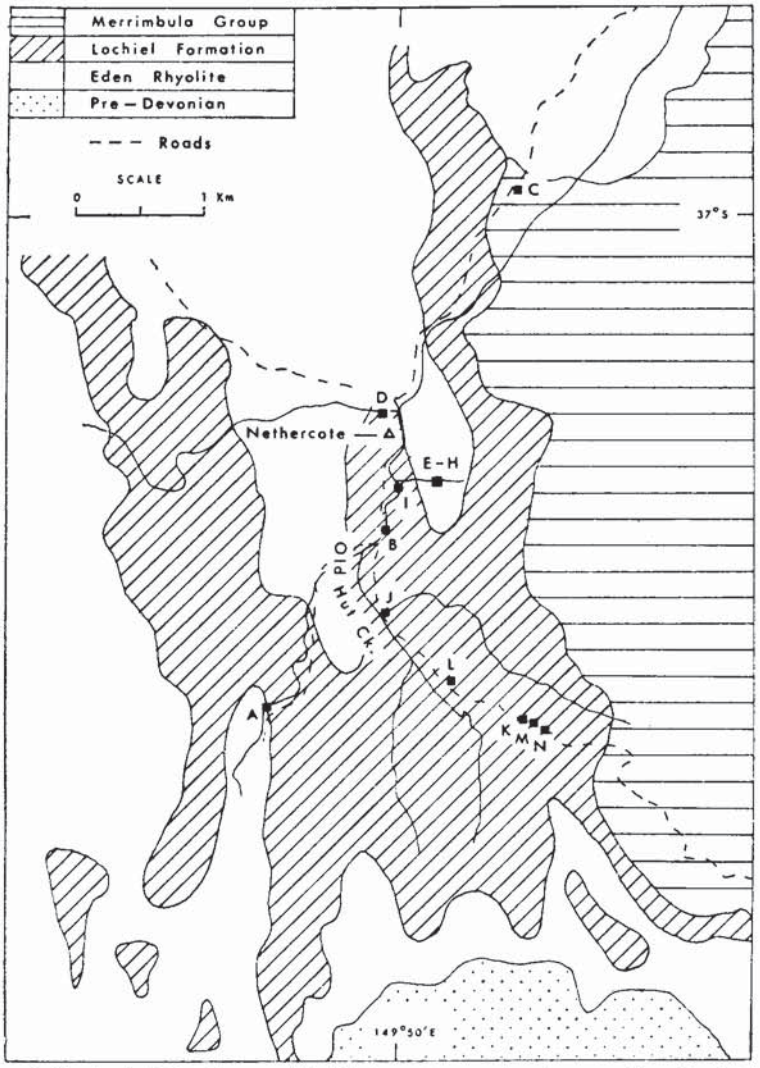


FIG. 3. Geological sketch map of the Nethercote area showing sampling sites in the Lochiel Formation and the Eden Rhyolite.

epidotization, are interbedded with arkoses, sandstones, red shales and conglomerates. These interbedded sediments are lithologically similar to the red beds of the conformably overlying Merrimbula Group, which is reliably assigned on palaeontological evidence to the Upper Devonian. The Lochiel Formation, with a thickness exceeding 400 m, is assumed to be also Late Devonian in age. The underlying Eden Rhyolite is taken as Middle Devonian. The rocks have been gently folded along meridional axes, with a slight regional tilting to the east.

3.2 *Sampling*

After tests on pilot samples from different parts of the outcrop, it was decided to make the new collection in the Nethercote area (Fig. 3). Here, the formation crops out along a valley where some flows are tilted as much as 30° (Steiner 1966).

Ten sites were located along the central creek and its tributaries. Five of these sites—C, E, F, G and H—were in the Eden Rhyolite. Another four sites, including the only one in sediments, were collected along the road that crosses a range of hills south of Nethercote. The 14 sites spanned a vertical distance of some 200 m.

3.3 *Laboratory measurements*

Nine samples remained from eight sites (G1 to G8) of Green's original collection, and an attempt was made to correlate these sites with the new ones. Two of Green's

Table 3

Lochiel Formation: accepted sample mean directions after cleaning

Site	Sample	*n*	*D*	*I*	*R*	*k*	Pole (deg)	
G3	E46	1	353	−35			130 E	71 N
G4	E48	2	06	−40	1·99	172	172 E	75 N
G5	E54	2	156	−08	2·00	1660	296 E	43 S
G6	E55	2	354	−22	1·99	102	137 E	64 N
G7	E58	2	158	21	2·00	1460	286 E	57 S
G8	E51	2	06	05	1·99	112	159 E	50 N
B	B1	2	23	−02	2·00	206	185 E	48 N
	B2	3	327	−20	2·99	194	94 E	50 N
	B3	2	330	—26	1·96	24	93 E	55 N
	B4	3	345	−09	2·99	209	123 E	55 N
D	D1	5	156	−11	4·93	57	297 E	42 S
	D2	3	151	11	2·96	49	283 E	49 S
	D3	4	138	—16	3·62	7·9	280 E	30 S
I	12	2	16	−11	2·00	1340	178 E	55 N
	I3	2	14	−10	2·00	2480	174 E	56 N
	I4	2	08	−18	2·00	685	167 E	61 N
	I5	2	07	−14	1·98	51	164 E	59 N
L	L1	2	178	18	2·00	736	325 E	62 S
	L2	2	175	26	1·99	133	317 E	66 S
	E43	3	168	09	3·00	1440	308 E	56 S
	E44	2	164	26	2·00	3840	294 E	62 S
M	M1	2	10	00	2·00	700	166 E	52 N
	M2	2	14	−01	2·00	1680	172 E	51 N
N	N1	2	19	12	2·00	5000	177 E	47 N
	N2	2	15	08	2·00	5000	171 E	47 N
Formation mean direction (α_{95} = 9°)								
		25	355	−11	23·0	12·0		
Formation mean of sample poles (A_{95} = 7°)								
					23·5	16·4	140 E	58 N

Intensities of NRM were in the range 1 to 700×10^{-6} emu cm^{-3}

Peak field normally used for cleaning was 500 Oe.

Symbols as in Table 2.

samples were assigned to site L and the results are reported accordingly (Table 3). The other samples could not reliably be referred to any of the recent sites. The directions of magnetization of five of these other seven samples were hardly affected by a.f. cleaning in peak fields of 500 Oe, but that of E54 moved to its ' reversed ' position from a direction much closer to the Earth's present field. The seventh sample, from site G7, failed to produce a stable direction although the other sample from this site possessed a very stable 'reversed' magnetization.

Seven of the new sites were collected from acid volcanics. The five sites in the Eden Rhyolite were rejected because specimens yielded directions of magnetization that were either unstable or very close to the present geomagnetic field. Both samples from site A were discarded for the same reasons. Four of the five samples from site B gave stable directions away from the Earth's field but the constituent specimens of the fifth, which was chemically altered, showed no within-sample agreement.

Six sites were collected from basic igneous rocks. The directions of NRM at site D were streaked between that of the present geomagnetic field and the presumed original

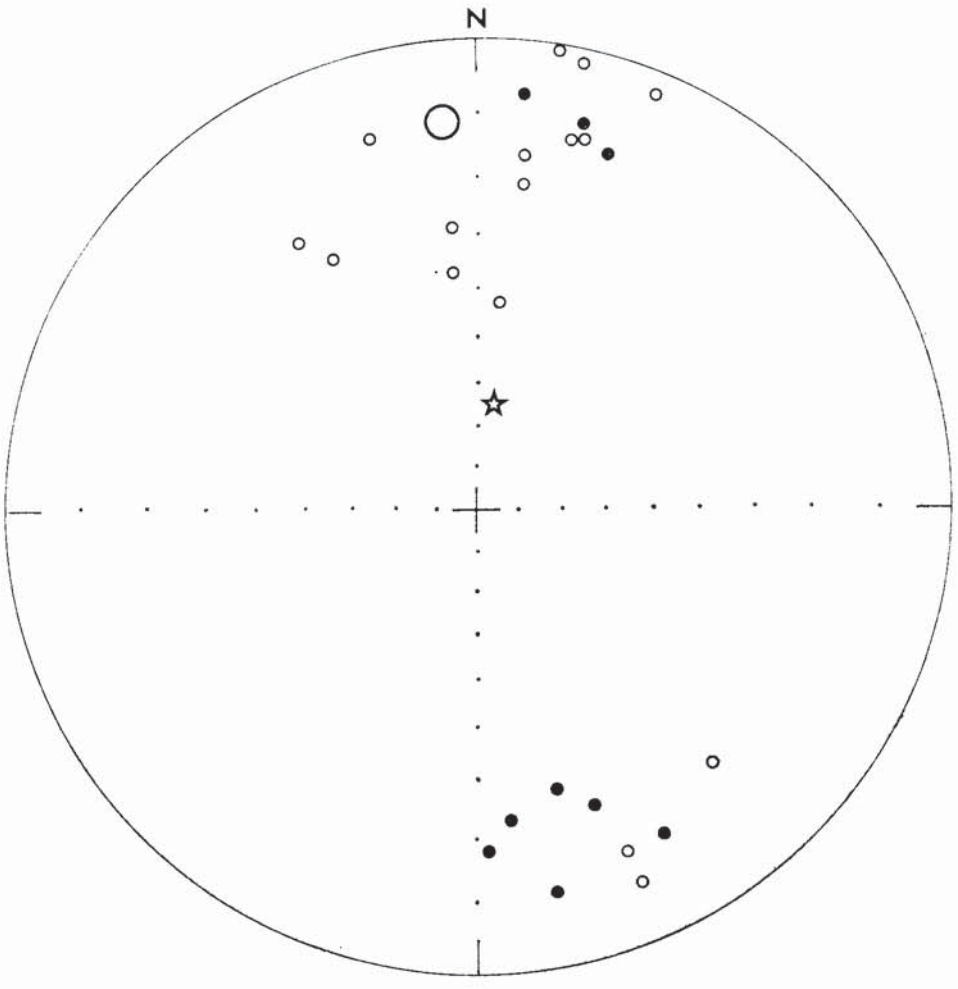


FIG. 4. Lochiel Formation. Mean cleaned directions for the 25 samples retaining the presumed original magnetization of the formation. Symbols are as in Fig. 2, the mean allowing for reversed magnetization of nine samples with directions in the 2nd quadrant.

' reversed ' magnetization revealed by magnetic washing. Samples from sites I, L, M and N generally demonstrated stable magnetizations, well away from the geomagnetic field, during a.f. treatment in peak fields as high as 1000 Oe. The magnetization of site J was also very stable—but in the direction of the Earth's present field. This site appeared to have undergone chemical alteration.

Site K, a red siltstone collected from a fresh road cutting, produced inconsistent results following thermal and a.f. demagnetization. The lack of between-sample agreement probably results from structural deformations of the sediment, which represents a zone of weakness in the formation. This explanation would also account for the anomalous dips at the location. The data were rejected.

3.4 *Results*

The results for an igneous formation are normally calculated from an analysis of site means. In the present case, however, not only are many sites represented by just one sample but it is also possible that several of Green's early sites have been unwittingly duplicated in the recent collection. Further, the possibility that the magnetization of sites B, D and L may have been recorded during more than a virtual instant of time is suggested by poor between-sample agreement. For these reasons, the formation mean direction of magnetization and the palaeomagnetic pole position (140°E, 58°N with $A_{95} = 7°$) have been computed from an analysis of sample means (Fig. 4, Table 3). A palaeomagnetic pole calculated from the twelve sites means would be situated at 140°E, 59°N, with $A_{95} = 10°$.

4. The Isismurra Formation (and Lower Kuttung)

4.1 *Geology*

The Carboniferous System is very well developed in the Hunter Valley of New South Wales, where a varied assortment of rock types records many changes in the prevailing environmental conditions.

A sequence of more than 3000 m of Lower Carboniferous rocks, including some volcanics of Viséan age, has been mapped in the Rouchel Area (Figs 1 and 5). The volcanics are incorporated in the Isismurra Formation as the Greenland Ignimbrite Member and associated tongues. Isolated outcrops of ignimbrite have also been correlated with the formation. Because of the extensive folding and faulting in the area, palaeomagnetic samples were collected only in localities where reliable structural control had been established.

4.2 *Sampling*

Hand samples were obtained from four separate localities (Fig. 5). Sites A and B were tongues of red ignimbrite exposed along a steep creek. Site C was a prominent cliff of andesite. Sites D and G were roadside exposures of two red tongues either side of site F, a blue tongue that forms a bar across Davis Creek. A single sample of sandstone, representing site E, was obtained close to D. Sites H to M were collected along Sandy Creek where it crosses the full width of the Greenland Ignimbrite Member.

4.3 *Laboratory measurements and results*

The directions of NRM varied considerably, with samples from sites C, F, H, L and M failing to show within-site agreement. The directions for A, B and K were within 10° of the Earth's present field, quite different from those for D, G and J—which were close to their respective cleaned results. Site E agreed with the adjacent site D, and site I produced a direction of NRM between that of the present geomagnetic field and that measured after cleaning.

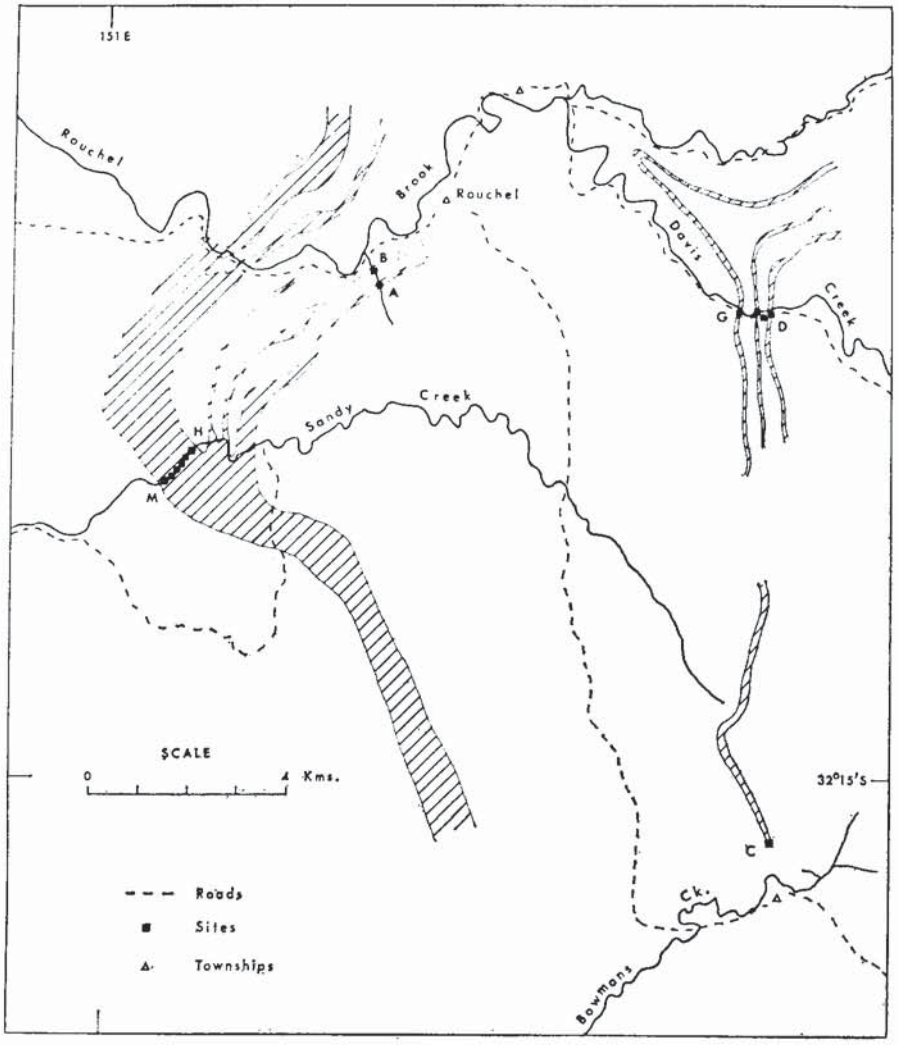


FIG. 5. Sketch map of the Rouchel area showing sampling sites and indicating the distribution of the Greenland Ignimbrite member and associated tongues of the Isismurra Formation.

The samples may be classified into three types, according to their different reactions to a.f. demagnetization.

1. Samples B1, B2, C1, K1, K2 and M1 failed to produce a stable magnetization.
2. Samples A1, A2. C2, C3, C4, E1, F1, F2 and H1 yielded apparently random directions of magnetization throughout the treatment.
3. Samples A3, D1, D2, G1, G2, G3, H2, H3, I1, J1, J2, J3, L1, L2 and M2 each revealed a stable direction of magnetization having good within-sample agreement.

Samples of the third type are considered to indicate an original magnetization, with the data from sites D, G, J and L having agreement that is good within the sites but relatively poor between the sites (Table 4). Samples H2 and H3 showed close

Table 4

Viséan Stage of the Hunter Valley: site mean directions after cleaning

Site	$N(n)$	D	I	R	k	Pole (deg)
			(a) Isismurra Formation			
A	1 (2)	10	−45	(1·98)	(41)	210 E 79 N
D	2 (4)	341	−13	1·98	62	113 E 59 N
G	3 (6)	205	79	2·99	180	317 E 50 N
H	2 (4)	188	70	2·00	544	318 E 68 N
I	1 (2)	249	−63	(1·99)	(98)	14 E 09 N
J	3 (6)	347	−62	2·95	41	08 E 75 N
L	2 (4)	22	−41	1·99	151	223 E 69 N
M	1 (4)	297	−34	(3·99)	(229)	59 E 32 N
			(b) Irving's Lower Kuttung			
H4	3 (7)	157	41	2·90		78 E 68 N
H5	2 (6)	147	37	1·96		74 E 59 N
H6	3 (8)	159	57	2·97		38 E 72 N
Viséan mean direction ($\alpha_{95} = 17°$)						
	11	340	−54	9·78	8·2	
Viséan mean of site poles ($A_{95} = 21°$)						
				9·26	5·7	34 E 73 N

Intensities of NRM were in the range 4 to 700×10^{-6} emu cm^{-3}.

Peak fields used for cleaning were 200–300 Oe.

Symbols as in Table 2.

agreement with one another and with site G—both sites being reversely magnetized with respect to the others. It is inferred that a class 3 sample produces a direction of original magnetization that is very similar to any other stable direction from the same site. The cleaned directions from samples A3, I1 and M2 are therefore believed to represent reasonably accurate spot readings of the ancient geomagnetic field. Each of the three samples has been accorded the status of a site in analysing the data.

A palaeomagnetic pole position calculated from the eight site results is associated with a value for A_{95} of 30°.

4.4 *Discussion*

It has previously been shown that ignimbrites can retain a stable magnetization over long periods of geologic time (e.g. Deutsch & Somayajulu 1970). However, the mode of formation of an ignimbrite is such that one might expect differences between sites, samples and perhaps specimens. Hence, the scatter of site directions and the inconsistent magnetic behaviour of samples might be attributable to the nature of the rock type. Nevertheless, it is noted that although the direction of magnetization of the very stable site D was the farthest from the formation mean, it agreed quite well with the NRM direction of an adjacent sedimentary sample (site E). This suggests that the scatter of site directions might be due to usually large short-term variations in the Viséan geomagnetic field.

Irving (1966) reported results from three sites in toscanites and felsites of the so-called Lower Kuttung Series which are also of Viséan age, as confirmed by K. S. W. Campbell (private communication). The directions of original magnetization, after correcting for the undoubted easterly tilt of 30° mentioned somewhat ambiguously by Irving, are quite similar to those recently obtained (Table 4, Fig. 6). Since only 60 km or so separates the two sampling areas, it seems reasonable to combine the results

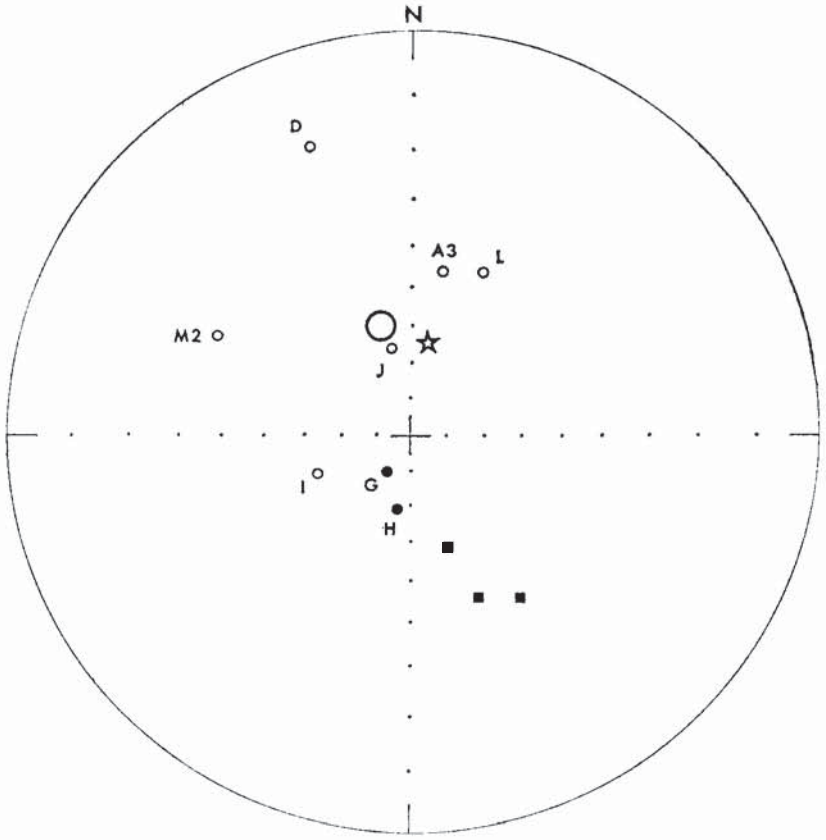


FIG. 6. Viséan stage of the Hunter Valley. Mean directions of magnetization after a.f. treatment are shown for eight sites in the Isismurra Formation (circles) and for Irving's three sites in the Lower Kuttung (squares). Symbols as in Fig. 2.

Table 5

Siluro—Devonian stratigraphy of the Yass District

Period	Stage	Rock Unit
Devonian	Emsian	Murrumbidgee ' Series '
	Siegenian	Bowning Group
	Gedinnian	
	Pridolian	Barambogie Group
Silurian	Ludlovian	Booroo Ponds Group
		Hattons Corner Group*
		Yass Group
	Wenlockian	Douro Volcanics
		Bango Beds
		Hawkins ' series '

* The basal member of this group is comprised of the volcanics of the Laidlaw Formation.

from all 11 sites in order to obtain a more meaningful palaeomagnetic pole position for the Viséan stage. The pole is situated at 34°E, 73°N with $A_{95} = 21°$.

5. Sampling in the Yass area

5.1 *Geology*

The Yass Basin is occupied by a Silurian sequence of alternating volcanic and non-volcanic units overlain by Devonian volcanics and sediments, the combined thickness being perhaps 4000 m. The succession has been described by Browne (1940, 1954) but a large area close to the town of Yass was recently mapped in detail by Link (1971) who updated the stratigraphy and correlated it with several other Australian and European sequences. Link's studies involved only Late Silurian and Early Devonian rocks, the terminology of Browne being used for other material listed in Table 5, which shows a simplified version of the stratigraphic succession.

The three Silurian volcanic sequences—the Hawkins ' Series ', the Douro Volcanics and the Laidlaw Formation—are very similar types, consisting largely of coarse tuffs, and are consequently difficult to distinguish in the field. Their ages are determined within close limits by the presence of abundant fossils in the interbedded sediments. The highest Silurian fossiliferous horizon occurs within the Barambogie Group, at the top of which are corals and fauna identified as early Gedinnian in age (Link 1970).

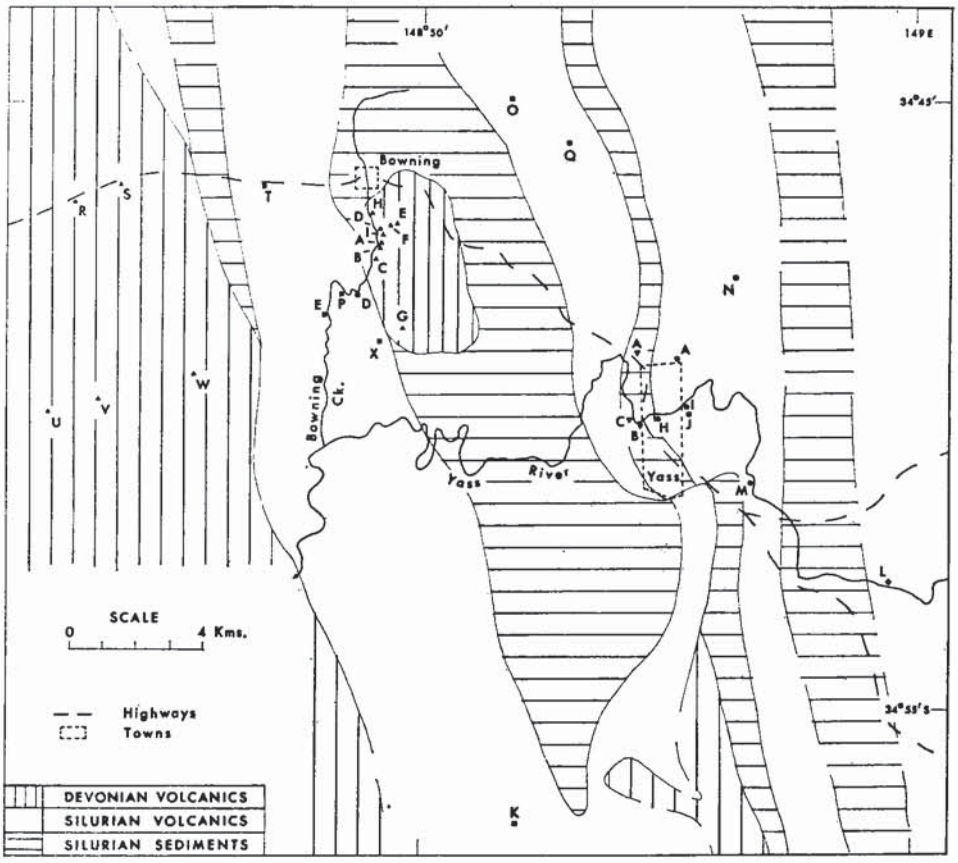


FIG. 7. Geological sketch map of the Yass district showing sampling sites in the Bowning Group (solid triangles), Yass Group (open triangles), Laidlaw Formation (squares), Douro Volcanics (circles) and Hawkins ' Series ' (diamond).

To the south-east of Bowning (Fig. 7), the volcanics of the Bowning Group consist of a basal andesite, an associated conglomerate, succeeding flows of rhyolite and an isolated occurrence of ignimbrite.

5.2 *Sampling*

Three sites were collected in sediments of the Yass Group. At both B and C (Fig. 7), four cores (50 cm × 2.8 cm) were drilled some 20 m apart in a grey calcareous quartz grit that was exposed in the bank of the Yass River. At A, two hand samples were obtained from a roadside exposure of blue, fine-grained sandstone.

The other 29 sites were in igneous rocks, and hand samples were collected from three types of exposure:—

1. Quarries in the Douro Volcanics:

Site J was a small quarry in massive rock some 25 m above the Yass River. Site A was a much larger quarry in very weathered rock capping a hill just north of Yass, but only one large block was obtainable from the rubble-covered working face.

2. Road Cuttings:

Sites R, S and W in the Bowning Group; K and T in the Laidlaw Formation; site N in the Douro Volcanics.

3. All other sites were collected from the banks of rivers or creeks.

5.3 *Laboratory measurements and sampling*

5.3.1 *The Bowning Group.* Intensities of NRM ranged from 10 to 1000×10^{-6} emu cm^{-3}. Sites R, S, U, V and W in the rhyolitic block to the west of the Yass Basin had not retained an identifiable original magnetization. Only sites V and W possessed good between-sample agreement of NRM, the direction being that of the Earth's present field. Use of the a.f. demagnetization method failed to change the directions from V and W, while samples from sites R, S and U showed no within-site agreement between their fluctuating directions of magnetization.

Of the other nine sites, only C, F and G possessed directions of NRM that were considered random. Magnetic washing revealed a stable primary magnetization for

Table 6

Bowning Group: site mean directions after cleaning

Site	$N(n)$	D		I	R	k	Pole (deg)	
A	5 (10)	38		−27	4·99	330	218 E	50 N
B	3 (6)	25		−45	3·00	443	224 E	67 N
C	2 (4)	30		−44	1·94	14	228 E	63 N
D	3 (6)	23		−33	2·98	97	201 E	67 N
E	2 (4)	37		−44	2·00	1080	233 E	58 N
F	2 (4)		**		1·34			
G	3 (6)		**		0·17			
H	6 (12)	21		−64	5·97	172	281 E	71 N
I	4 (8)	19		−35	3·91	34	201 E	67 N
Formation mean direction ($\alpha_{95} = 10°$)								
	7	28		−42	6·84	36·8		
Formation mean of site poles ($A_{95} = 9°$)								
					6·86	42·3	225 E	64 N

Peak field normally used for cleaning was 200–300 Oe.

Symbols as in Table 2.

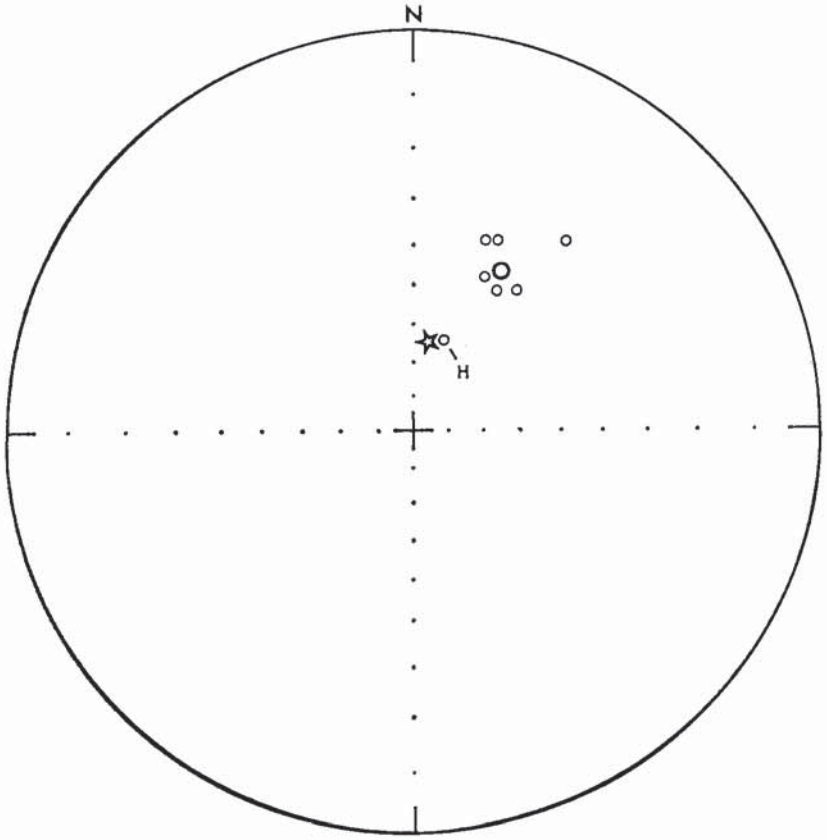


FIG. 8. Bowning Group. Mean cleaned directions are shown for the seven sites with non-random magnetization after a.f. treatment.

sites A, B, C, D, E, H and I. This magnetization is presumed to be original, for the reasons given below, results being listed in Table 6 and illustrated in Fig. 8. Although the magnetization of site H appears to be directed along the present geomagnetic field, this is after correcting for a dip of 25° at the site. The seven sites with usable results spanned a vertical height of 100 m. Site G was a conglomerate containing cobbles of Gedinnian andesite that had eroded before the rhyolite flows of late Gedinnian and Siegenian times. Specimens from three samples showed good within-sample agreement of directions of NRM. The mean direction for the site possessed a vector resultant ' R ' of only 1·80 which does not indicate non-randomness by the criterion of Vincenz & Bruckshaw (1960). Magnetic cleaning methods hardly affected the magnetization of the samples, although the value of ' R ' tended to decrease. The random directions of magnetization of the samples indicate that the conglomerate has not been uniformly magnetized since the Early Devonian. The stability of the directions during treatment suggests that the cobbles are retaining their original magnetization. This evidence, similar to the conglomerate *test of Graham (1949),* implies that the stable magnetization observed in the andesitic sites is probably original.

A Gedinnian palaeomagnetic pole position calculated from the data is at 225°E, 64°N with $A_{95} = 9°$.

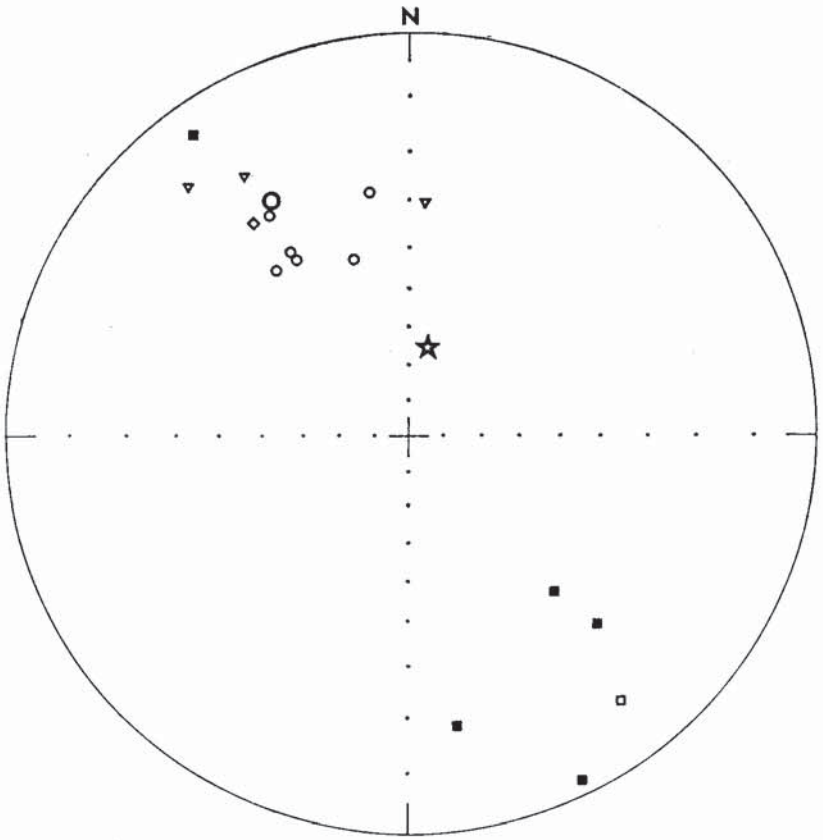


FIG. 9 Silurian rocks. Mean cleaned directions for 13 sites in volcanics and directions of NRM for one sample and two cores from sediments of the Yass Group. Symbols as in Figs 2 and 7.

5.3.2 *The Silurian rocks.* The sediments were very weakly magnetized but specimens from one hand-sample and two cores possessed NRMs considered to be non-random. The directions remained unchanged to peak fields of 200 Oe, after which the magnetization was too weak for measurement. The results are shown in Fig. 9, which indicates the directions of presumed original magnetization for the Silurian sites.

The intensities of NRM of the igneous rocks were generally in the range 30 to 500×10^{-6} emu cm^{-3}. However, two of the samples from site L in the Hawkins 'Series' were magnetically too weak at $0{\cdot}2 \times 10^{-6}$ emu cm^{-3} for a.f. demagnetization. Their directions of NRM agreed well with those from the other two samples from the site, which were not affected by peak alternating fields of 500 Oe.

The directions of NRM of all samples in the Douro Volcanics were closely grouped and stable in peak alternating fields as high as 500 Oe. No 'reversed' directions were noted.

The samples of the Laidlaw Formation behaved similarly to those of the Douro Volcanics. Exceptions were site K, whose samples were very weakly magnetized, and site O, whose samples showed no within-site agreement. Sites D, P, Q, T and X are reversely magnetized with respect to other Silurian samples.

Mean directions of magnetization for the Douro Volcanics and Laidlaw Formation are shown in Table 7. The substantial overlap of their circles of 95 per cent confidence

Table 7

Silurian Volcanics: site mean direction after cleaning

Site	$N(n)$	D		I	R	k	Pole (deg)	
				(*a*) Laidlaw Formation				
LF D	2 (4)	153		03	2·00	2260	97 E	48 N
LF E	2 (4)	324		04	1·99	96	99 E	40 N
LF K	4 (8)		**					
LF O	3 (6)		**					
LF P	4 (8)	141		−09	3·89	27	98 E	36 N
LF Q	2 (4)	135		22	2·00	216	77 E	43 N
LF T	2 (4)	168		09	1·98	41	126 E	58 N
LF X	3 (6)	137		33	2·87	16	71 E	48 N
Mean direction (α_{95} = 18°)+								
	6	327		−11	5·67	15·4		
				(b) Douro Volcanics				
DV A	1 (4)	351		−27	3·98	186	124 E	68 N
DV H	3 (6)	328		−33	2·99	155	80 E	57 N
DV I	6 (12)	321		−35	5·96	111	73 E	53 N
DV J	4 (8)	328		−34	3·99	229	80 E	57 N
DV M	2 (4)	327		−24	2·00	1460	87 E	52 N
DV N	2 (4)	343		−40	1·99	151	93 E	71 N
Mean direction (α_{95} = 10°)								
	6	333		−33	5·90	49·3		
				(c) Hawkins ' series '				
HS L	2 (4)	324		−23	2·00	219	85 E	50 N
Wenlockian/Ludlovian mean direction (α_{95} = 10°)								
	13	329		−22	12·35	18·4		
Wenlockian/Ludlovian mean of site poles (A_{95} = 7°)								
	13				12·63	32·5	91 E	54 N

+ Mean obtained after reversing magnetization of 5 sites.
Peak fields used for cleaning were in range 200–500 Oe.
Symbols as in Table 2.

indicates that the two results are indistinguishable at this level of confidence. The data for all the Silurian volcanics have been combined and the resultant palaeomagnetic pole position for Middle and Upper Silurian is at 91°E, 54°N with A_{95} = 7°.

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